

```
user$ ip addr show
```

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: tunl0@NONE: <NOARP> mtu 1480 qdisc noop state DOWN qlen 1000
    link/ipip 0.0.0.0 brd 0.0.0.0
3: ip6tnl0@NONE: <NOARP> mtu 1452 qdisc noop state DOWN qlen 1000
    link/tunnel6 00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00 brd 00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00
6: ipvlan-eth0@tunl0: <BROADCAST,MULTICAST,UP,LOWER_UP,M-DOWN> mtu 65000 qdisc noqueue state UNKNOWN
    link/ether 42:00:a9:fe:01:02 brd ff:ff:ff:ff:ff:ff
    inet 169.254.8.1/16 brd 169.254.255.255 scope global ipvlan-eth0
        valid_lft forever preferred_lft forever
    inet6 fddf:3978:feb1:d745::c001/64 scope global flags 02
        valid_lft forever preferred_lft forever
    inet6 fe80::4200:a900:1fe:102/64 scope link
        valid_lft forever preferred_lft forever
7: ipvlan-eth1@ip6tnl0: <BROADCAST,MULTICAST,UP,LOWER_UP,M-DOWN> mtu 65535 qdisc noqueue state UNKNOWN
    link/ether 42:00:4e:49:43:00 brd ff:ff:ff:ff:ff:ff
```

```
user$ netstat -an
```

Active Internet connections (servers and established)

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State
tcp	0	0	0.0.0.0:8080	0.0.0.0:*	LISTEN
tcp	0	0	169.254.169.1:8080	169.254.169.126:11878	ESTABLISHED
tcp	0	0	fddf:3978:feb1:d745::c001:44462	2001:4860:4841:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:45784	2001:4860:4844:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:57126	2001:4860:4843:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:52236	2001:4860:4841:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:40718	2001:4860:4845:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:42486	2001:4860:4847:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:43124	2001:4860:4847:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:56348	2001:4860:4841:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:46388	2001:4860:4846:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:33748	2001:4860:4847:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:47054	2001:4860:4843:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:39520	2001:4860:4842:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:35140	2001:4860:4843:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:51158	2001:4860:4843:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:40718	2001:4860:4844:400:::443	TIME_WAIT
tcp	0	0	fddf:3978:feb1:d745::c001:59504	2001:4860:4844:400:::443	TIME_WAIT

```
tcp      0      0 fddf:3978:feb1:d745::c001:38302 2001:4860:4842:400:::443 TIME_WAIT
tcp      0      0 fddf:3978:feb1:d745::c001:46398 2001:4860:4846:400:::443 TIME_WAIT
tcp      0      0 fddf:3978:feb1:d745::c001:48200 2001:4860:4844:400:::443 TIME_WAIT
tcp      0      0 fddf:3978:feb1:d745::c001:42498 2001:4860:4847:400:::443 TIME_WAIT
tcp      0      0 fddf:3978:feb1:d745::c001:52244 2001:4860:4841:400:::443 TIME_WAIT
```

Active UNIX domain sockets (servers and established)

Proto	RefCnt	Flags	Type	State	I-Mode	Path
unix	3	[]	STREAM	CONNECTED	1163	
unix	3	[]	STREAM	CONNECTED	1162	
unix	3	[]	STREAM	CONNECTED	158504	
unix	3	[]	STREAM	CONNECTED	158503	
unix	3	[]	STREAM	CONNECTED	3379	
unix	3	[]	STREAM	CONNECTED	3380	
unix	3	[]	STREAM	CONNECTED	155328	
unix	3	[]	STREAM	CONNECTED	155327	
unix	3	[]	STREAM	CONNECTED	936	
unix	3	[]	STREAM	CONNECTED	937	

user\$ hostname

localhost

```
user$ nslookup google.com
```

```
Server:          169.254.169.254
```

```
Address:         169.254.169.254:53
```

```
Non-authoritative answer:
```

```
Name:   google.com
```

```
Address: 74.125.126.102
```

```
Name:   google.com
```

```
Address: 74.125.126.101
```

```
Name:   google.com
```

```
Address: 74.125.126.100
```

```
Name:   google.com
```

```
Address: 74.125.126.139
```

```
Name:   google.com
```

```
Address: 74.125.126.138
```

```
Name:   google.com
```

```
Address: 74.125.126.113
```

```
Non-authoritative answer:
```

```
Name:   google.com
```

```
Address: 2607:f8b0:4001:c62::65
```

```
user$ arp -a
```

```
? (169.254.169.126) at 42:01:a9:fe:a9:7e [ether] on ipvlan-eth1
```

```
? (169.254.1.254) at 42:00:a9:fe:01:fe [ether] on ipvlan-eth0
```

```
user$ ip neigh show
```

```
169.254.169.126 dev ipvlan-eth1 lladdr 42:01:a9:fe:a9:7e ref 1 used 0/0/0 probes 4 REACHABLE
169.254.1.254 dev ipvlan-eth0 lladdr 42:00:a9:fe:01:fe ref 1 used 0/0/0 probes 1 REACHABLE
fddf:3978:feb1:d745::c002 dev ipvlan-eth0 lladdr 42:01:a9:fe:01:01 router ref 1 used 0/0/0 probes 1 REACHABLE
```

```
user$ route -n
```

```
Kernel IP routing table
```

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Iface
0.0.0.0	169.254.1.254	0.0.0.0	UG	0	0	0	ipvlan-eth0
169.254.0.0	169.254.1.254	255.255.0.0	UG	0	0	0	ipvlan-eth0
169.254.1.0	0.0.0.0	255.255.255.0	U	0	0	0	ipvlan-eth0
169.254.9.0	0.0.0.0	255.255.255.0	U	0	0	0	ipvlan-eth1
169.254.169.0	169.254.169.126	255.255.255.128	UG	0	0	0	ipvlan-eth1

```
user$ ip route show
```

```
default via 169.254.1.254 dev ipvlan-eth0
169.254.0.0/16 via 169.254.1.254 dev ipvlan-eth0 src 169.254.8.1
169.254.1.0/24 dev ipvlan-eth0 scope link
169.254.9.0/24 dev ipvlan-eth1 scope link src 169.254.9.1
169.254.169.0/25 via 169.254.169.126 dev ipvlan-eth1 src 169.254.169.1
```

```
user$ curl https://example.com
```

```
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:,"><meta name="viewport"
content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-
serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This
domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a
href="https://iana.org/domains/example">Learn more</a></p></div></body></html>
```

```
user$ wget https://example.com/file.zip
```

```
[admin@localhost ~]$ ethtool eth0
netlink error: no device matches name (offset 24)
netlink error: No such device
netlink error: no device matches name (offset 24)
netlink error: No such device
netlink error: Operation not permitted
netlink error: no device matches name (offset 24)
netlink error: No such device
netlink error: no device matches name (offset 24)
netlink error: No such device
netlink error: no device matches name (offset 24)
netlink error: No such device
netlink error: no device matches name (offset 24)
netlink error: No such device
No data available
[admin@localhost ~]$ nmcli device status
DEVICE    TYPE        STATE                CONNECTION
enp0s3    ethernet    connected            enp0s3
lo        loopback    connected (externally) lo
[admin@localhost ~]$ sudo systemctl restart
[sudo] password for admin:
admin is not in the sudoers file.  This incident will be reported.
[admin@localhost ~]$ █
```